Artefact Take Home Assignment

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This report summarizes the steps taken to clean and transform the "Customer Reviews" data set, extract sentiment and themes from each review, build EDA charts, and provide high level recommendations

The complete analysis, including code, visualizations, and intermediate outputs, is documented in a Jupyter Notebook available on GitHub:

GitHub Repository: https://github.com/jadkrisht98/Artefact_Project/blob/main/Final_Notebook.ipynb

1. **Methodology:**

1.1 Data Transformation & Preprocessing:

The dataset underwent several preprocessing steps to ensure data quality and analytical reliability. These steps included:

- **Handled missing values** — identified rows with null ratings (whose content was also unclear) and dropped them.
- **Removed duplicates** — checked for and dropped duplicate rows (ignoring the tags column, since the same review can have different tags).
- **Removed emoji-only reviews** — found rows where the review content was nothing but emojis (no actual text) and dropped them, since they carry no analyzable information.
- **Extracted tag details** — split each tag dictionary into separate columns (tag, sentiment) and merged with the JSON lookup table to add offering and destination columns.
- **Extracted rating details** — split each rating dictionary into normalized_rating and actual_rating columns.
- **One-hot encoded offerings** — converted the offering column into four binary (0/1) flag columns, one per offering category. Then grouped back to one row per review, so each review shows which offerings it relates to.
- **Resolved multiple destinations** — when a review was linked to multiple destinations (from multiple tags), kept only the most frequently occurring destination for that review.
- **Re-normalized all ratings to a 0–100 scale** — applied a simple rule: multiply the actual rating by 20 (so 1→20, 2→40, ..., 5→100), and cap anything above 5 at 100. This ensures a consistent 0–100 normalized scale regardless of the original rating system.
  Sample data frame is snap below:

	id	content	date	language	title	normalized_rating	actual_rating	sentiment	offering_Accommodation	offering_Food & Beverage	offering_Retail	offering_Tourism Attractions/ Sites	destination
0	377380-167052110186	جميل جدن خدمه فاخره مميزه	2021-02-07 00:14:00+00:00	ara	Hotel Raffles Makkah Palace	100.0	5.0	missing	1	0	0	1	Makkah
1	377380-167053227344	مكان جميل في شرما على البحر	2021-02-07 00:01:00+00:00	ara	HYAT SHARMA RESORT	60.0	3.0	missing	1	0	0	0	Tabuk
2	377380-167056314181	Good Hotel	2021-02-07 01:51:00+00:00	eng	Novotel Yanbu	80.0	4.0	missing	1	0	0	0	Yanbu
3	377380-167057432860	منتجع جميل يحتاج إلى ترتيب وأهتمام	2021-02-07 00:45:00+00:00	ara	منتجع و سبا درة الرياض / Durrat Al Riyadh Reso...	80.0	4.0	missing	1	1	0	0	Riyadh
4	377380-167058489640	جيده الاستقبال محترمين	2021-02-07 02:28:00+00:00	ara	ديم للغرف الفندقية	60.0	3.0	missing	1	0	0	0	Dammam
...	...	...	...	...	...	...	...	...	...	...	...	...	...
9946	377380-208857650522	Nice and good	2021-06-01 12:07:00+00:00	eng	Burger King - Riyadh Gallery 2	100.0	5.0	missing	1	1	0	0	Riyadh
9947	377380-225738358384	رائعه وجميله جدا	2022-10-17 11:16:00+00:00	ara	Prince Hussam Park	100.0	5.0	missing	0	0	0	1	Al Baha
9948	377380-259913223515	أجمل متنزهات السعودية فعلاً رائعة	2023-10-03 06:31:00+00:00	ara	Arruddaf Park	100.0	5.0	missing	0	0	0	1	Taif
9949	377380-259913232905	جميييله جدا ورائعه	2023-10-03 02:30:00+00:00	ara	Arruddaf Park	100.0	5.0	missing	0	0	0	1	Taif
9950	377380-259913238900	المتنزه الرئيسي بمدينة الطائف يقصده السكان وال...	2023-10-01 11:28:00+00:00	ara	Arruddaf Park	100.0	5.0	missing	0	0	0	1	Taif

9951 rows × 13 columns

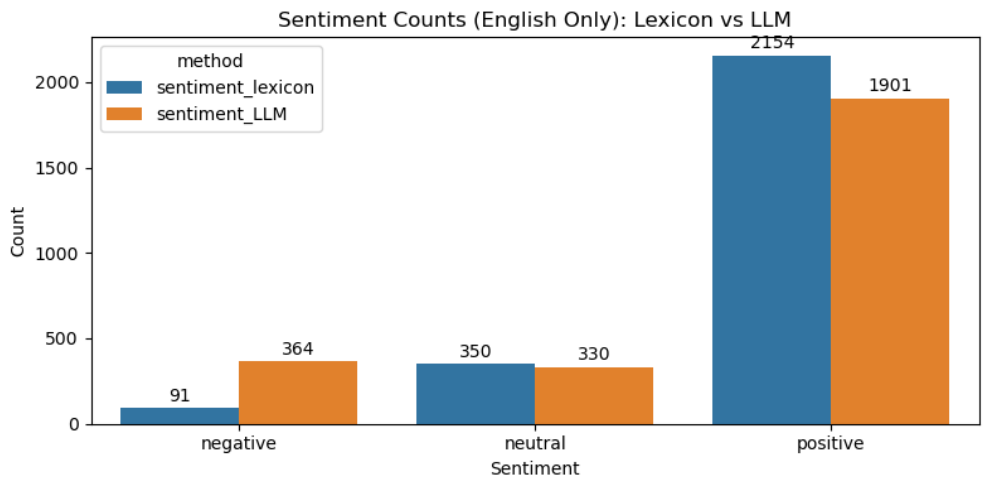
1.2 Sentiment Analysis:

I conducted sentiment analysis on customer reviews using two distinct methods: (1) a lexicon-based approach, and (2) a deep learning technique utilizing a pre-trained large language model (LLM).

For the lexicon-based method, I employed the **TextBlob** library, which assigns sentiment scores to individual words in the text and calculates an overall sentiment for each review. This approach is straightforward and works well for English-language content, but unfortunately, I could not find an equivalent library for Arabic text, limiting its use to English reviews only.

For the deep learning approach, I utilized a task-specific LLM model called **twitter-XLM-roBERTa** (link), which is designed for multilingual sentiment analysis. This model consists of two main components: the XLM-RoBERTa pre-trained LLM that generates contextual word embedding, and an additional neural network layer that classifies each review as positive, neutral, or negative.

The model is integrated via the **tweetnlp** library (link), enabling robust sentiment classification across multiple languages, including both English and Arabic reviews.



I compared the 2 models (English reviews only) manually by looking at 20 samples. The LLM based approach is much more reliable and precise. A sample is attached below:

```
#Manually comparison when 2 mehods disagree for English reviews only
final_df[(final_df['language'] == 'eng') & (final_df['sentiment_LLM'] != final_df['sentiment_l
```

	content	sentiment_LLM	sentiment_lexicon
293	More(Translated by Google) Muslim kiss and smeared patch	neutral	positive
296	Wow, world 1st ranked review place,	positive	neutral
299	World peaceful place where every Muslim wants to go	neutral	positive
313	Hopefully my stay will be long-term. Stayed in a few differer	positive	neutral
336	Food was alright. The chicken in the pasta was fried very we	positive	neutral
338	Better to buy tickets to Santorini and visit Oia in person	neutral	positive
387	Need to improve its ambience overall ok mall	negative	positive
400	They have apple bee's we celebrated our wedding anniversa	neutral	positive
409	View is good, furniture is nice. Recpition organization and se	negative	positive
425	Lazy housekeeping .. no car parking service .. you can hear t	negative	positive

1.3 Text Cleaning:

The following steps were taken to clean up the text:

- **Translation handling**: Reviews that include both original Arabic and Google-translated English are split to keep only the translated English segment; the language is forced to English for those rows.
- **Language validation**: A rule-based checker (Arabic vs English character detection) flags mismatches and updates the language column when confidently identified as English or Arabic.
- **English cleaning**: Non-letter characters are removed, text is lowercased, English stopwords are removed, and tokens are lemmatized.
- **Arabic cleaning**: Diacritics are removed, non-Arabic characters are stripped, text is tokenized with an Arabic tokenizer, and Arabic stopwords are removed.

Output: A cleaned field clean_content is created and used for downstream NLP; the original content column is dropped afterward.

1.4 Theme/topic extraction:

For deriving themes out of the reviews, I used 2 methods, (1) Topic Modeling via **LDA** (Latent Dirichlet Allocation) - classic probabilistic approach, and (2) Ensemble ML model – **BERTopic**.

For method (1) I trained an LDA model with 5 topics, listed the top words per topic, then manually named each topic as a theme. For each review/content, I selected the most likely topic and stored its theme label in the dataset.

Topic 1	جدا, جميل, مكان, الله, ممتاز, فندق, المكان, الفندق, رائع, جيد, نظيف, الاستقبال, يوجد, ونظيف	Accommodation & Positive Venue Experience
Topic 2	place, good, nice, beautiful, one, best, restaurant, food, great, visit, service, family, excellent, park	Food & Dining / Restaurants
Topic 3	ممتاز, رائع, god, mosque, جميلة, may, prayer, حديقة, peace, جدا, تحتاج, جميله, oh, upon	Religious / Spiritual & Tranquil Sites
Topic 4	افضل, الرياض, سوق, يوجد, مكان, مول, يحتاج, أجمل, جميلة, او, جيده, حديقة, المملكة, شي	City Amenities & Shopping (Riyadh-focused)
Topic 5	جميله, شي, المكان, قلعة, اللي, اجمل, مارد, الفندق, علي, القلعة, بقاع, الأرض, نصيف, الملك	Attractions & Scenic / Heritage Sites

Review count per theme:

Accommodation & Positive Venue Experience	4596
Food & Dining / Restaurants	2550
Religious / Spiritual & Tranquil Sites	1106
City Amenities & Shopping (Riyadh-focused)	985
Attractions & Scenic / Heritage Sites	714

However, method (1) is very manual to label and limited as it doesn't consider the contextual meaning of the word (same word could have different meaning in different reviews). So, **alternatively I considered a more advanced methodology - BERTopic.**

For method (2). I used an ensemble model /pipeline called "BERTopic" (link to repo) created by Maarten Grootendorst. In summary it's a topic-modeling method that clusters document embeddings and extracts topic keywords with TF-IDF.

BERTopic is a pipeline that combines multiple steps:

1. Create vector embeddings via BERT
2. Cluster reviews via HDBSCAN
3. TF-IDF keyword extraction (top 10)
4. Optional Representation model to name themes (I used OpenAI gpt-4.1 model)

I chose the model to produce 20 themes/topic, below is a snapshot from the code with final topic names and count. Please refer to GitHub repo for full detailed code.

```python
#ADDING TOPIC REPRESENTATIONS LAYER
#Use chat-gpt from open AI to fine tune topic representations instead of manually coming up with

import openai
from bertopic.representation import OpenAI
import os

openai_api_key = os.environ['OPENAI_API_KEY']
client = openai.OpenAI(api_key=openai_api_key)

# Update topic representations using OpenAI
system_prompt="'You are an assistant that extracts high-level topics in english from customer re
representation_model = OpenAI(client=client, model="gpt-4.1", system_prompt=system_prompt)

texts = final_df['clean_content'].astype(str).tolist()

topic_model.update_topics(docs=texts, representation_model=representation_model)

topic_info = topic_model.get_topic_info()
topic_info[['Topic', 'Name', 'Count', 'Representation']]
```

✓ 10.0s Python

Topic	Name	Count	Representation
-1	-1_Positive experiences	30	['Positive experiences']
0	0_Beautiful places	1786	['Beautiful places']
1	1_Positive customer feedback	1770	['Positive customer feedback']
2	2_Religious expressions	1366	['Religious expressions']
3	3_Hotel cleanliness quality	1261	['Hotel cleanliness quality']
4	4_Restaurant dining experience	879	['Restaurant dining experience']
5	5_Need for improvement	876	['Need for improvement']
6	6_Shopping malls	525	['Shopping malls']
7	7_Temporarily closed businesses	369	['Temporarily closed businesses']
8	8_Peaceful and Beautiful Atmospher	204	['Peaceful and Beautiful Atmosphere

Count	Representation
192	['Family-friendly place']
96	['Fresh fish quality']
133	['Bathroom cleanliness issues']
97	['Parking and Access']
82	['Cinema experience']
63	['Customer satisfaction']
47	['Star rating feedback']
50	['Nighttime park experience']
48	['Catalina plane wreck']
77	['Tourist attractions']

Count	Representation
30	['Positive experiences']
1786	['Beautiful places']
1770	['Positive customer feedback']
1366	['Religious expressions']
1261	['Hotel cleanliness quality']
879	['Restaurant dining experience']
876	['Need for improvement']
525	['Shopping malls']
369	['Temporarily closed businesses']
204	['Peaceful and Beautiful Atmosphere

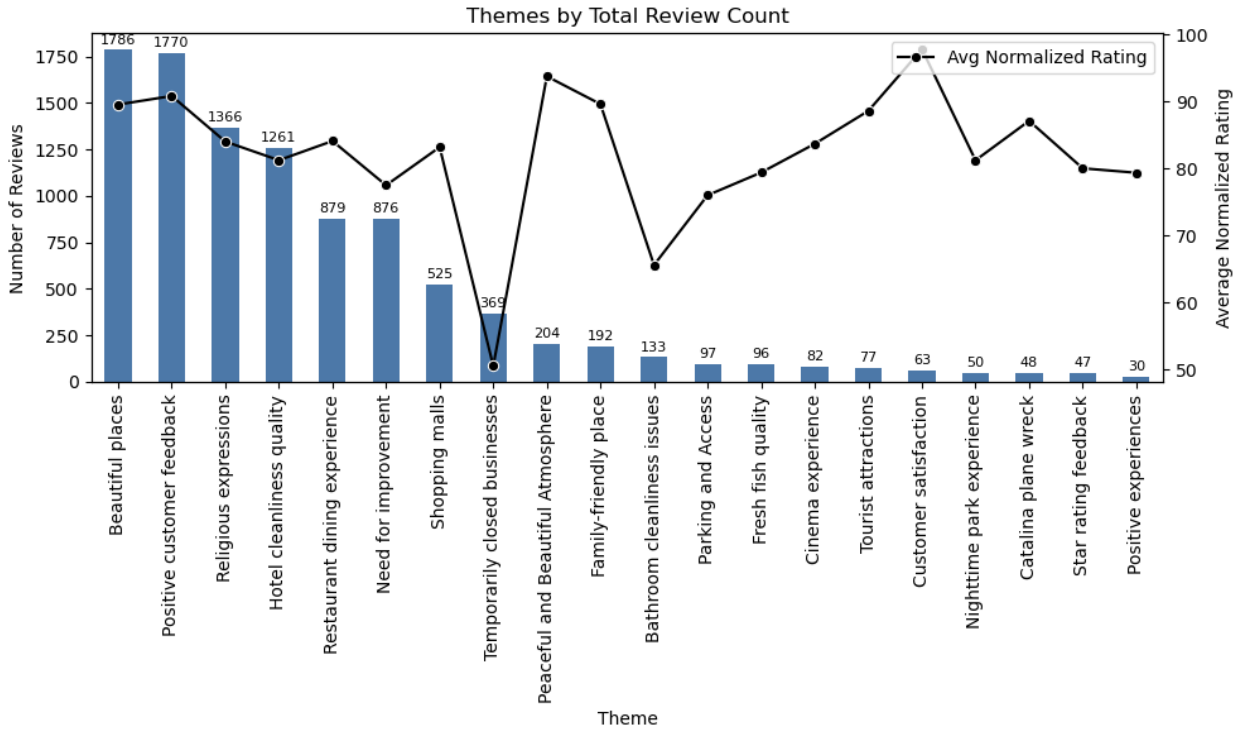
The data is now clean, with sentiment analysis and topic extraction complete—ready for EDA and model training/testing.

2. **Findings:**

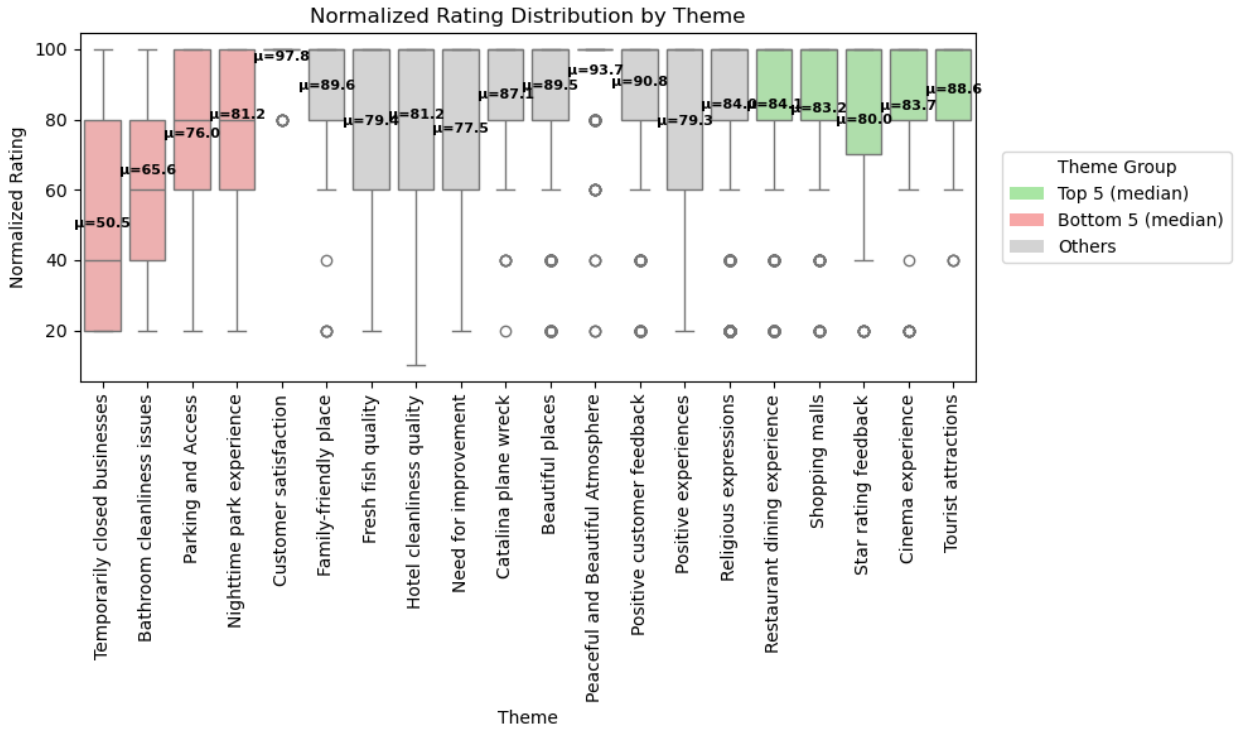
2.1 Insights from Common Keywords and Themes:

Top 20 common words used in the reviews based on TF-IDF scores are:

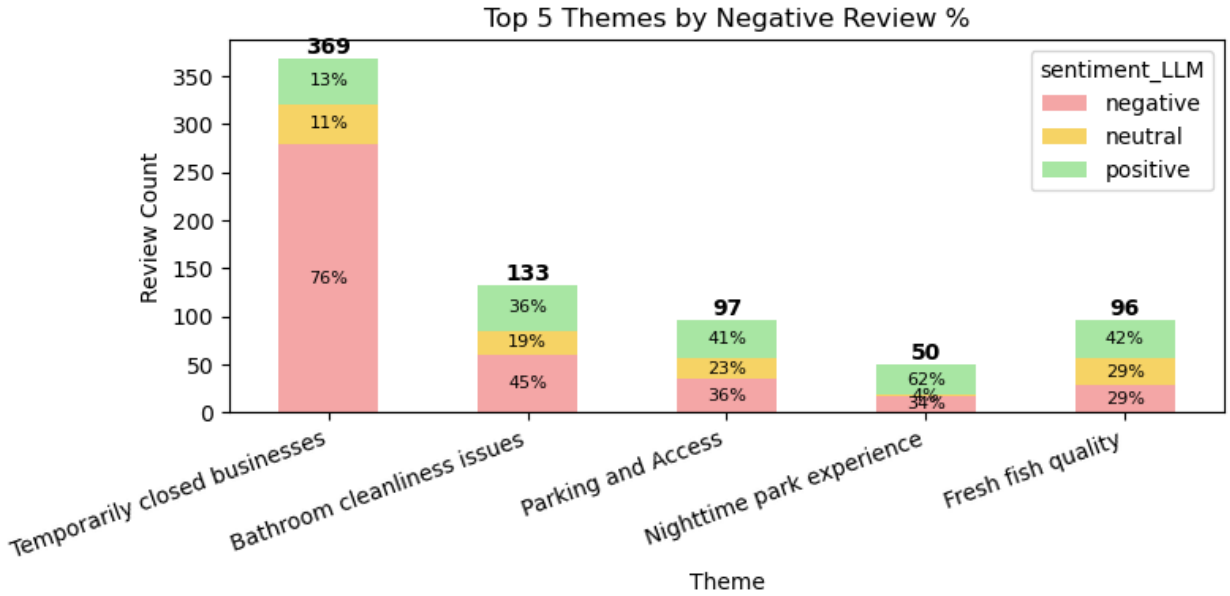
With مكان, رائع, ممتاز, جدا, جميل being in top 5



Review volume is concentrated on a small number of themes: Beautiful places, Positive Customer Feedback, Religious Expressions, Hotel cleanliness quality.

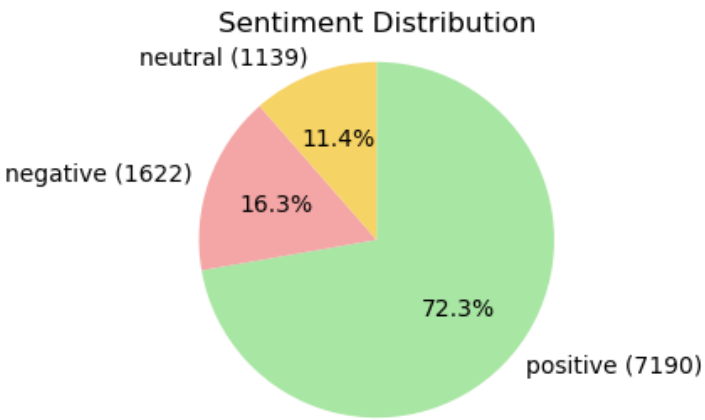


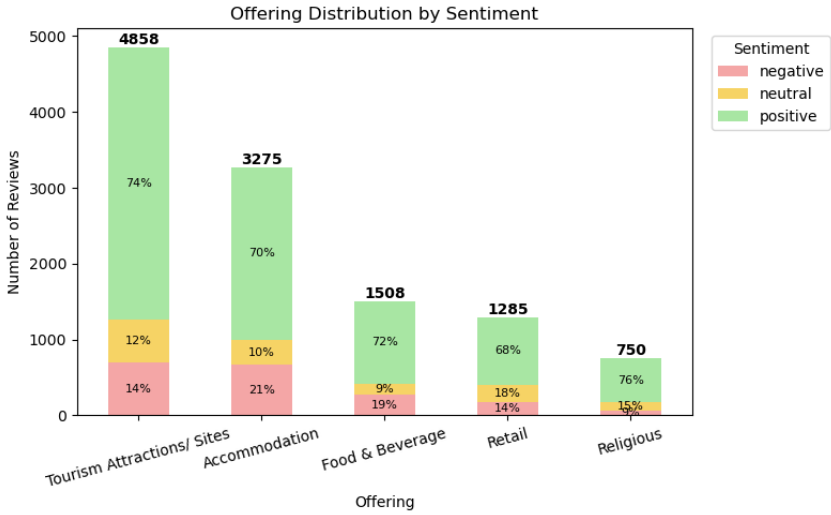
Themes with the lowest average normalized rating: **Temporarily closed businesses**, **Bathroom cleanliness issues**, **Parking and Access**, **Nighttime Park experience**, and **Customer Satisfaction**



Negative reviews are most common for themes: **temporarily closed businesses**, **bathroom cleanliness issues**, **parking and access**, **night time park experiences**, and **fresh fish quality**.

2.2 Insights from Sentiment Analysis:





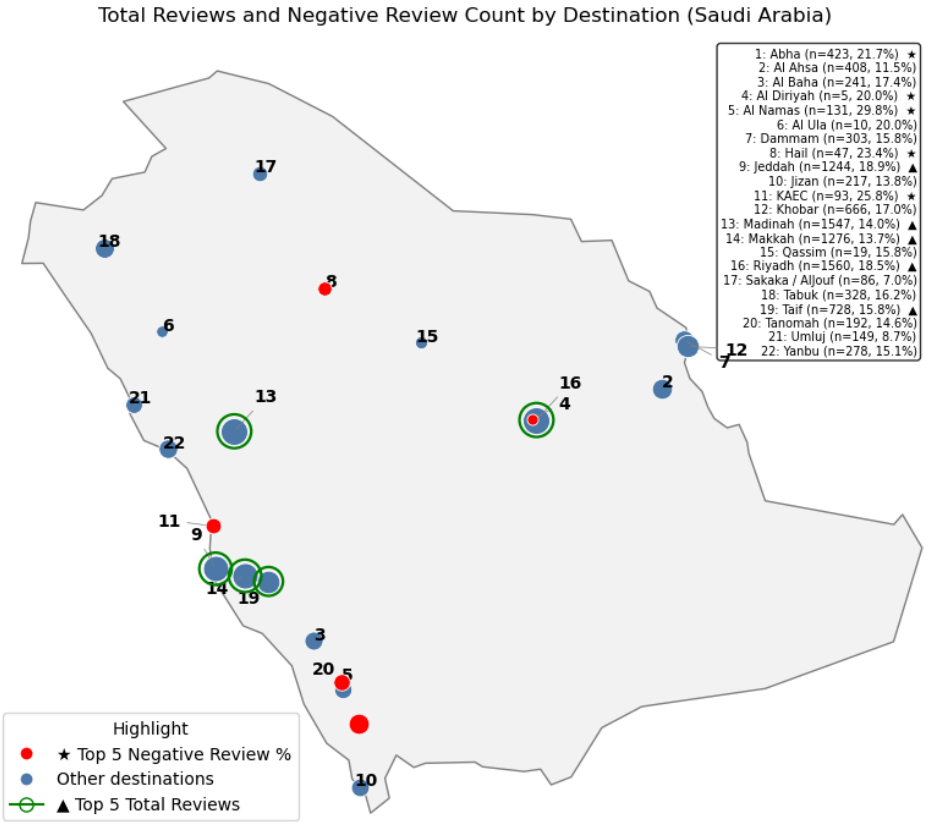
Customer feedback is mostly positive (72%). However, looking into the negative reviews alone, places with offering **Accommodations** are on top of the list in term of % of negative reviews after it comes **Food & Beverages**.

Looking at different destinations in KSA, the ones with the most negative reviews % are:

Abha (22%), Al Diriyha (20%), Al Namas (30%), Hail (23%), and KAEC (26%)

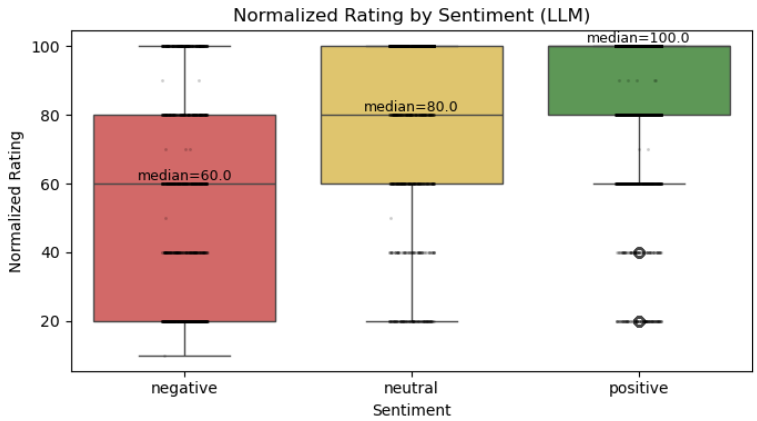
The destinations with the most reviews are:

Taif (728), Riyadh (1560), Makkah (1267), Jeddah (1244), Madinah (1547)



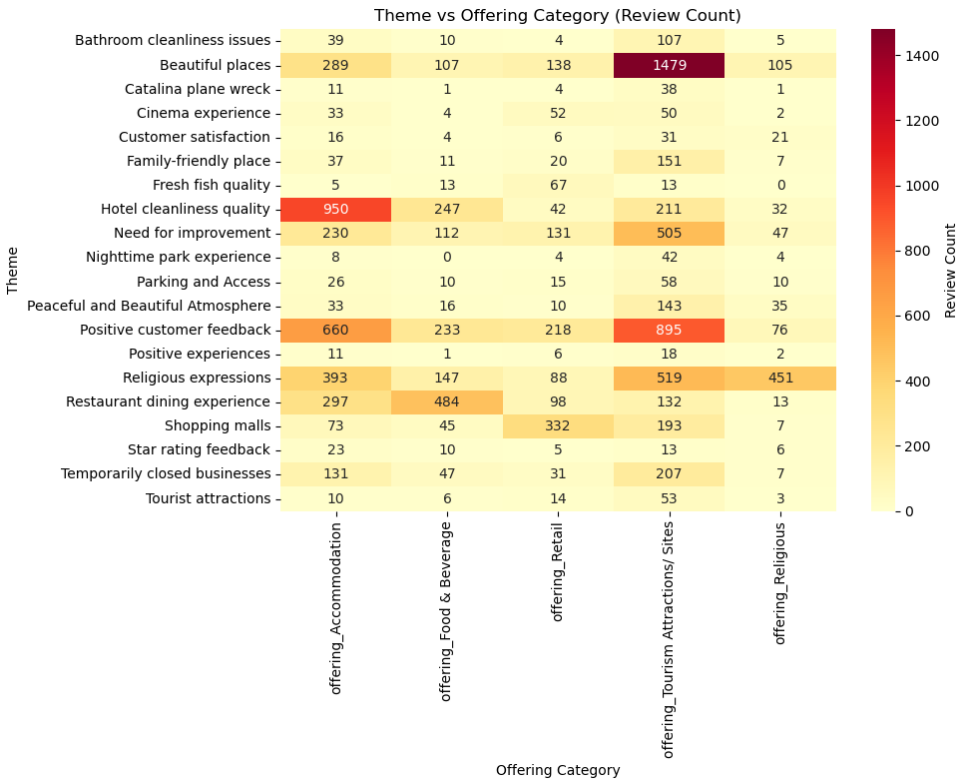
2.3 Other insights from EDA:

The distribution of normalized ratings by sentiment shows a clear monotonic relationship: the higher the rating the more positive the sentiment.



The "Theme vs Offering Category (Review Count)" heatmap indicates that themes are not uniformly distributed across offerings; instead, certain themes concentrate heavily within specific offering types.

This is analytically useful because it shows where to act: if a high-negative theme is concentrated in a particular offering category, interventions can be scoped precisely.



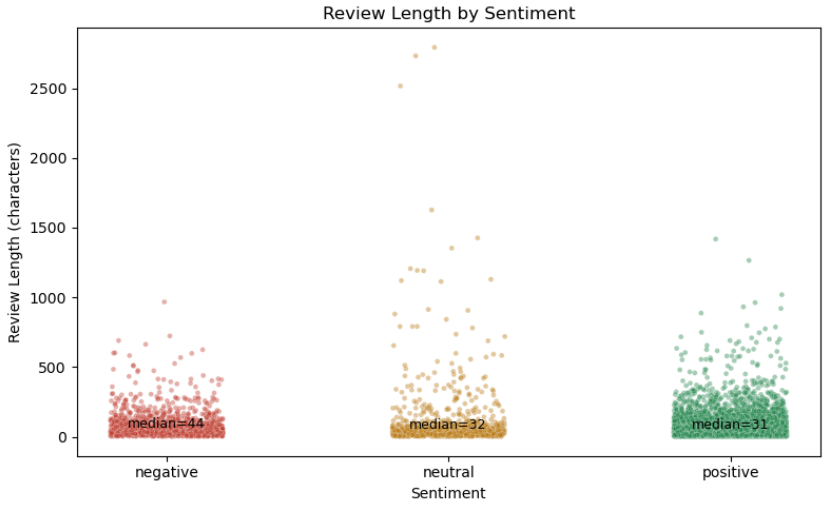
Beautiful places → Tourism attraction site

Hotel Cleanliness quality → Accommodation

Positive Customer feedback → Tourism attraction site

The median length of negative reviews is 44 characters, while positive reviews typically have a median length of just 31 characters.

This pattern suggests that longer reviews are generally linked to lower ratings. Customers who are dissatisfied or have complaints tend to write more detailed feedback, often elaborating on specific issues or concerns. In contrast, positive reviews are usually shorter, reflecting straightforward satisfaction without the need for extensive explanation.



3. **Recommendation:**

Issue	Action	Value
High-impact negative themes	Verify status, enforce hygiene standards, improve access	Resolves key customer pain points.
Negative sentiment in Accommodation and Food & Beverages	Improve service quality and staff training	Reduces negative reviews and boosts reputation.
Strong positive themes	Promote in marketing and bundles	Attracts visitors and reinforces strengths.
Destination-specific negativity	Run localized audits	Maximizes impact with targeted fixes.
Themes concentrated by offering	Fix issues where they cluster	Enables focused, efficient improvements.
Long negative reviews	Flag for review and follow-up	Early detection of dissatisfaction.

4. Future Scope:

Develop an NLP classification model for sensitivity analysis that categorizes reviews as positive, neutral, or negative.
The classification will be based on key aspects compiled so far, including destination, offering, theme, and normalized rating. This targeted approach will help capture nuanced sentiments across different dimensions.
To evaluate model performance, implement both multi-class logistic regression and random forest algorithms, then compare their accuracy and weighted F1 scores. Because the dataset is imbalanced, with a majority of positive reviews, it is crucial to:

- Stratify the target variable when splitting the data, ensuring that minority classes are equally represented in both the training and testing datasets.
- Use macro F1 score and recall as primary evaluation metrics, prioritizing the detection of negative reviews since false negatives are particularly costly.
- Apply higher penalties to misclassifications of minority classes, preventing the model from defaulting to minority class predictions. (set class_weight='balanced' while fitting model)